VF Asia Footwear Laboratory Testing Manual

Test Method for Whole Shoe Flex	Test method # VF- G02 Page 1 of 5
Date Issued : November , 2016 **Revision # : 4** **Reason for revision :** 1. Changed the flexing angle based on category 2. Replaced support block from wood to PU Foam Foot. 3. Modify the word on marking -" if using an insole as reference is NOT practical, use the outsole ".	**Created by : Nila Babia Erazo**

PURPOSE AND SCOPE: To determine the ability of a shoe to withstand repeated flexing / bending to a 60 degree angle. The method simulates the flexing which a shoe undergoes when the wearer is walking or running.

REFERENCES: SATRA TM 92

EQUIPMENT AND MATERIAL:

1. Whole Shoe flexing machine.
2. Support block : Soft PU LAST
3. Vernier caliper (graduated to 0.02 mm)
4. Assembled Shoe

SAMPLE PREPARATION:

1. Condition the test sample in a standard controlled environment at 23 +/- 2 °C and at 55 +/- 5 % RH for at least 48 hours to allow the cement to crosslink completely.

2. Draw a line down the length of the insole of the shoe from the center of the heel towards the center of the toecap. Note: ***if using an insole as reference is NOT practical, use the outsole.***

3. Measure along the line the approximate distance given in **table 1**, from the heel end of the line and mark a point to the nearest millimeter

4. Mark the second point on the line drawn which is 5 ± 1 mm closer to the heel.

5. Draw a line across the width of the insole board or ***outsole*** which passes through the second point and at 90 ° to the line drawn. This line is regarded as the flexing line of the shoe / footwear. ***Refer to figure 1.***

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Table 1:

Children's	**Men's** **Note: for Women's should use 1 size lower. If fitting is still too tight may use 2 size lower.** ***Ensure that the flexed area of PU Foam Foot aligned with the marked flexing line of the shoe / footwear.***

Size	Heel / ball length (on insole or O/S for Vulcanize shoes) mm	Size	Heel /ball length (on insole or O/S for Vulcanize shoes) mm
4	85	1	143
5	91	2	149
6	97	3	155
7	102	4	160
8	108	5	166
9	114	6	172
10	120	7	178
11	126	8	184
12	131	9	189
13	137	10	195
		11	201
		12	207

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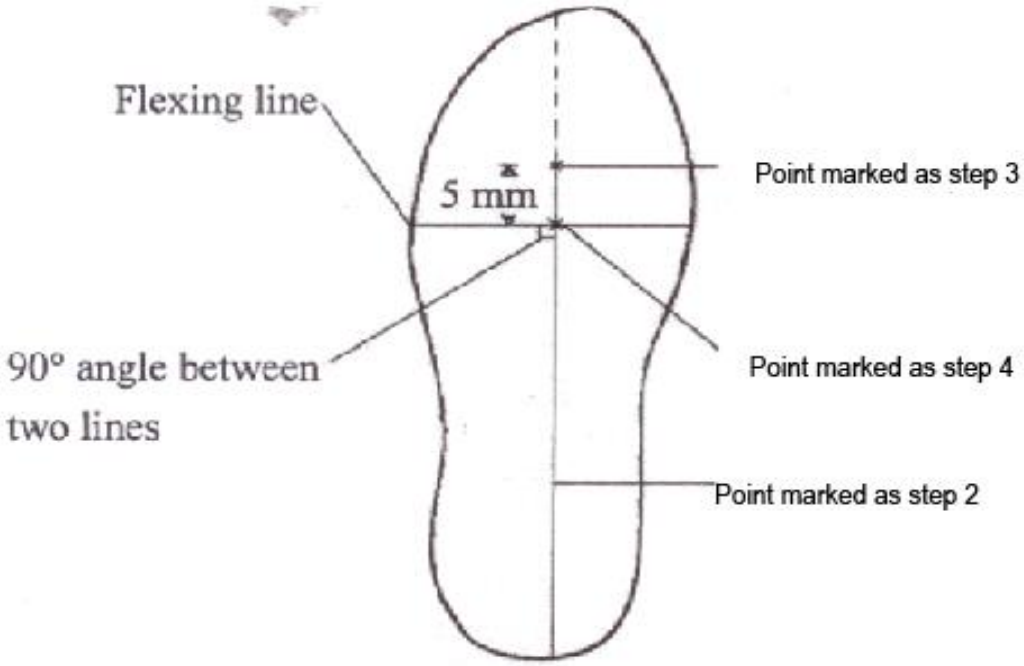


Figure 1

6. Do not remove the sock-liner.

7. Lace and tie the shoe in a normal manner. (Lace opening should typically be about 3 cm or any acceptable opening if the style doesn't permit).

REPETITIONS: Half or a pair of assembled footwear.

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TEST PROCEDURE:

A) **Test at atmospheric temperature @ 23±2 °C:**

1. Open the side protective cover on the Whole shoe flex machine, and set the bending angle for the following type of footwear:
   ***Option 1:** angle @ 60 degrees for running shoes*
   ***Option 2:** angle @ 45 degrees for vulcanize, casual shoes*
   ***Option 3:** angle @ 30 degrees for thick sole such as PRO mining, smelter or logger boots*

2. Place the PU Foam Foot into each shoe to be tested – same size as the shoe sample. ***Note:** If the actual size indicated on the PU Foam Foot is too tight, may use the 1 size lower.*

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Figure 2

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3. Position the shoe on the "Whole Shoe Flexing Machine "such that the flex area is located at the joint of the machine base and the flexing bar. Ensure that the edge of the round block is on the flexed area. ***. Refer to figure 2.***

4. With the flexed area properly positioned, tighten the heel clamp firmly onto the support block.

5. Secure the front of the shoe with the toe clamps.

6. Set the "shut –off" counter to the appropriate number of cycles.

7. Set the speed at 130±10 rev/min

8. Set the cycle counter to zero.

9. Initiate testing.

10. Check the shoes at least every 5,000 cycles over the first 25,000 cycles, after 25,000 cycles, check the shoes at least every 10,000 cycles.

11. Record the point at which any of the followings:

- Delamination of the outsole to midsole bond
- Delamination of the midsole to upper bond
- Any other delamination in the flex area
- Cracking or other damage to the midsole, outsole or upper materials.

B) Test at freezing temperature – 20°C or at specified freezing temperature.

Follow steps in section A, Dry Test at 23+/-2°C .The difference is that the test is carried out in the freezing chamber.

Note: Ensure that the chamber should be closed and allowed to reach @ - 20°C temperature or **at the specified freezing temperature**, before the start of the test.

REPORT: 1.The number of cycles at which damage to the flex area of the shoe first occurs.

2. The length and depth of delamination at the standard required cycles in mm.